Session 5 (DATA): Exercise sheet

Foundations of Clinical Research · Level 1 · Neudata · *#ClearDataClearImpact* · Week 5 · Saturday 28 November 2026 · online, 19:30–22:30

Name: _____ **Department:** _____

This sheet is sent by email before the session, as a Word file and as a PDF. Type your answers into the Word file (or print the PDF). In class we do Exercises 2, 6 and 7 together as a whole class (you answer in the chat or a poll; the instructor types the class answer on the shared screen). The other exercises are homework or self-check. No calculator is needed: every exercise is about *choosing* and *interpreting*. Exercises marked **[FLUID-ICU]** use the course dataset FLUID-ICU (simulated) and the results packs in `Course/Data/results_packs/` (Results_Pack_Q1 to Q5, also sent as PDFs). Exercise 8 becomes part of your own proposal (Part 3).

#	Exercise	When	Time
1	Classify the variables	homework / self-check (after the 5.1 quiz)	5 min
2	Fix the bad CRF items	5.1 whole-class task	7 min
3	Choose the right summary	5.3 check / homework	3 min
4	Judge a bad figure	homework	5 min
5	[FLUID-ICU] Spot the problems in the raw FLUID-ICU data	homework (after the 5.2 demo)	5 min
6	[FLUID-ICU] Read the FLUID-ICU Table 1	5.4 whole-class task	10 min
7	[FLUID-ICU] Your manuscript question as a PECO sentence	5.2 whole-class task (Q1); later for the question you choose	4 min
8	Proposal builder 5 worksheet (your own proposal)	proposal builder: instructor models, you draft; finish as homework	30 min

Exercise 1: Classify the variables

For each variable, tick **one** type.

#	Variable (ICU study of septic shock)	Continuous	Count	Binary	Nominal	Ordinal	Time-to-event
a	Age at ICU admission (years)	☐	☐	☐	☐	☐	☐
b	Sex (male / female)	☐	☐	☐	☐	☐	☐
c	Glasgow Coma Scale on admission (3–15)	☐	☐	☐	☐	☐	☐
d	Main source of sepsis (lung / urine / abdomen / skin / other)	☐	☐	☐	☐	☐	☐
e	Number of vasopressors running at 24 h (0, 1, 2, 3)	☐	☐	☐	☐	☐	☐
f	Cumulative fluid balance at day 5 (ml/kg)	☐	☐	☐	☐	☐	☐
g	Alive at day 28 (yes / no)	☐	☐	☐	☐	☐	☐
h	Time from ICU admission to death (patients alive at day 28 are censored)	☐	☐	☐	☐	☐	☐
i	Pain: none / mild / moderate / severe	☐	☐	☐	☐	☐	☐
j	Blood group (A, B, AB, O)	☐	☐	☐	☐	☐	☐

Bonus: which ONE of these would you most like to collect as an exact number rather than in groups, and why?

Exercise 2: Fix the bad CRF items

In class the instructor shows one item at a time on the shared screen; type your rewrite in the chat. Each item below can be filled in two different ways by two different nurses. Rewrite it so that it has **only one possible reading**. Check yourself: unit? time point? one answer only? an "unknown / not done" option?

#	Bad item	What is wrong?	Your rewritten item
a	Fluid given: _____		
b	Weight: _____		
c	Comorbidities: _____ ___		
d	Lactate high? Y / N		

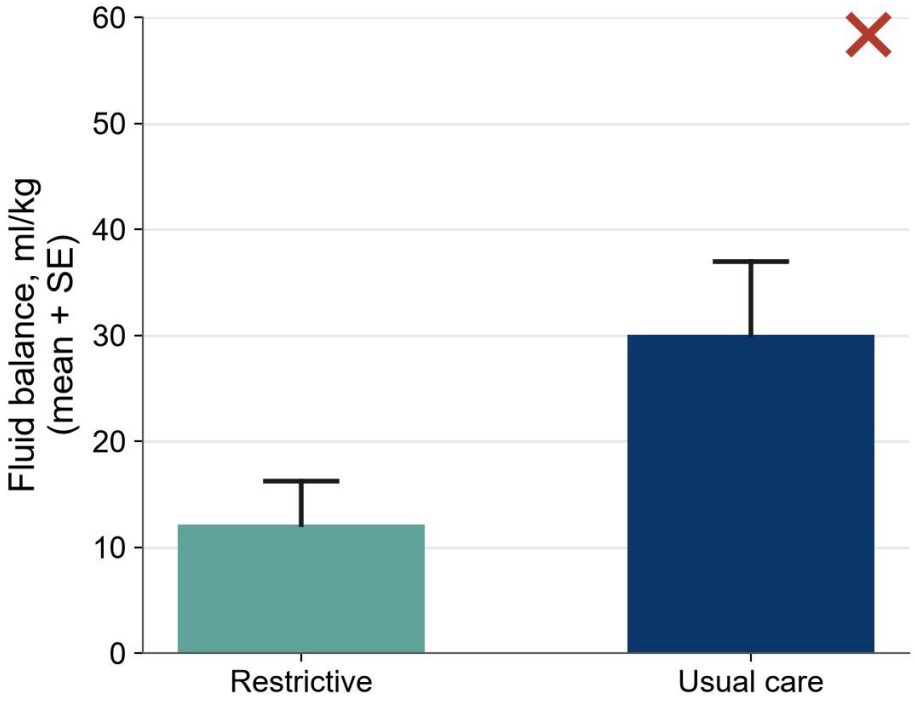
Exercise 3: Choose the right summary

For each variable, choose the summary you would report in Table 1 and the best graph.

#	Variable (200 ICU patients)	Shape / type	Summary	Graph
a	Age (years), bell-shaped			
b	ICU length of stay (days): most 1–5 days, a few > 30 days			
c	Died in hospital (yes / no)			
d	C-reactive protein (mg/L): long right tail			
e	Source of sepsis (5 categories)			
f	Systolic blood pressure, compared between two treatment groups			

Exercise 4: Judge a bad figure

A poster in the hospital corridor shows this figure (fluid balance at day 5, 24 patients per group):



1. What does each bar show, and what does the error bar show? What is missing?
2. Name **two things the reader cannot see** in this figure.
3. Redraw it (sketch) the way you would show it in a paper. What do you gain?

A second poster shows ward mortality as bars of 31% (Ward A) and 29% (Ward B), with the y-axis running from 28% to 32%.

4. What impression does it give, and how would you fix it?
5. A third poster puts hand-hygiene compliance (%) and infection rate on one graph with two different y-axes and writes "hygiene works!". Give two reasons to be careful.

Exercise 5: [FLUID-ICU] Spot the problems in the raw FLUID-ICU data

Below are six cells copied from the raw file `Course/Data/fluid_icu_raw.csv` (as the four ICUs sent it). For each, say what is wrong and what the cleaning script should do.

#	Column	Value in the raw file	What is wrong?	What should the script do?
a	sex	`m`, `male`, `Female`		
b	admission date	`14/03/2023` in one row, `14-Mar-2023` in another		
c	lactate (ICU C)	`23.4`		
d	lactate	`999`		
e	age	`210`		
f	patient ID	`FLU-139` appears in two identical rows		

Bonus: why must the raw file never be edited by hand? Name two things the cleaning log records for every change.

Exercise 6: [FLUID-ICU] Read the FLUID-ICU Table 1 (results pack)

Open Table 1 (baseline characteristics by fluid-balance group) in any results pack: the same table is in all five (`Course/Data/results_packs/Results_Pack_Q1` to `Q5`, PDF). In class we answer questions 1 to 5 together; question 6 is for you.

1. How many patients are in each column? Why is the overall number 618 and not 620?
2. For **three** variables, write the summary used (mean (SD), median (IQR) or n (%)) and why.

Variable	Summary used	Why this summary?

3. Were the two groups alike at the start? Name the **biggest** difference between the negative- and positive-balance groups.
4. Which variable has the most missing values, and how many?

5. Write **two sentences** describing the patients, as they would appear at the start of the Results section of your manuscript (numbers in brackets, no p-values).

6. Looking at this table, would you expect the crude association between positive fluid balance and death to be larger or smaller than the adjusted one? Why?

Exercise 7: [FLUID-ICU] Your manuscript question as a PECO sentence (5.2)

In class we fill this in together for Question 1 (Q1). Afterwards, fill it in for the ONE question (Q1 to Q5) you choose for your manuscript. PECO: P = population, E = exposure, C = comparison, O = outcome.

	Your answer
Your chosen question (Q1 to Q5), copied from page 1 of its results pack	
P: population	
E: exposure (variable name in the data dictionary) and its type	
C: comparator	
O: outcome (variable name) and its type	
One-sentence research question	
Which test will compare the groups? Which model will adjust? (guess now, check in Session 6)	

Exercise 8: Proposal builder 5 worksheet (individual)

The instructor models each part on the running example; then you draft the variables table in class (8 minutes) and finish the rest as homework. Write your answers here first, then copy them into Part 3 of your own proposal template (`Course/Templates/Research_Proposal_Template.docx`). (Your **own** question, not the course dataset.)

8.1 Variables table (8 min)

Role	Variable	How measured (unit, time point)	Type
Primary outcome			
Secondary outcome (optional)			
Exposure / intervention			
Confounder 1			
Confounder 2			
Confounder 3			
Descriptive (e.g. age, sex)			

Tip: if your design is an RCT, confounders are less of a problem (randomisation), but list the key baseline variables you will show in Table 1.

8.2 Mini-CRF: 6–10 items (10 min)

#	Question exactly as the data collector will read it	Answer format (boxes / ticks / unit)	"Unknown / not done" option?
1			☐
2			☐
3			☐
4			☐
5			☐
6			☐
7			☐
8			☐

8.3 Data dictionary: 3–5 rows + one data-quality rule (6 min)

Variable name	Label (meaning)	Type	Unit / codes	Allowed values

Data-quality plan (one or two sentences): who checks the forms, how often, and what range/consistency check you will run.

8.4 Data-management plan (6 min)

Question	Our answer
Paper CRF with double data entry, or electronic data capture (e.g. REDCap / KoBoToolbox)?	
Coded study IDs? Where is the linking log kept, and who can open it?	
Validation rules at entry (two examples from your data dictionary)	
Audit trail / cleaning log: who records changes?	
Backups (3-2-1) and data-sharing statement in the consent form	